

Exercise - Time Complexity

Solve these with respect to best case and worst case and specify where best case and worst case is same or not distinguishable:

1. `bool isPrime(int n) {`

`for (i = 2; i < n; ++i)`

`if (n % i === 0)`

`return false;`

`return true;`

`}`

2. `bool isPrime(int n) {`

`for (i = 2; i < sqrt(n); ++i)`

`if (n % i === 0)`

`return false;`

`return true;`

`}`

3. `void show(int n){`

`for (i = 1; i <= n; ++i)`

`cout << i << ' ';`

`}`

4. `void show(int n){`

`for (i = 1; i <= n; i = i / 2)`

`cout << i << ' ';`

`}`

5. `void show(int n){`

`for (i = 1; i <= n; i++)`

`for (j = 1; j <= n; j++)`

`cout << i*j << ' ';`

`}`

```
6. void show(int n){
    for (i = 1; i <= n; i++)
        for (j = 1; j <=n;j=j+2)
            cout << i*j << ' ';
}
```

```
7. void show(int n){
    for (i = 1; i <= n; i++)
        for (j = 1;j <=n;j=j*2)
            cout << i*j << ' ';
}
```

```
8. void show(int n){
    for (i = 1; i <= n; i++)
        for (j = 1;j <=i;j++)
            cout << i*j << ' ';
}
```

```
9. void show(int n){
    for (i = 1; i <= n; i++)
        for (j = i;j <=n;j++)
            cout << i*j << ' ';
}
```

```
10. void show(int n){
    for (i = 1; i <= n; i=i*2)
        for (j = 1;j <=n;j++)
            cout << i*j << ' ';
}
```